NET732

Week 6:Host Intrusion Analysis

Practical

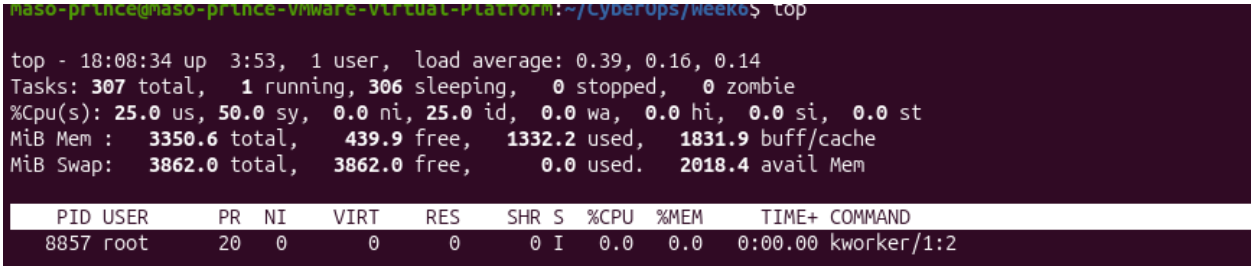
1. Checking running processes

I used Ubuntu Terminal to check the processes running on my computer.

ps aux

I also used:

top




These commands helped me view running programs and check their resource usage.

2. Checking running services

I used the following command:

systemctl list-units --type=service --state=running

```
maso-prince@maso-prince-VMware-Virtual-Platform:~/CyberOps/Week6$ systemctl list-units --type=service --state=running
  UNIT                            LOAD   ACTIVE SUB     DESCRIPTION
  accounts-daemon.service         loaded active running Accounts Service
  avahi-daemon.service            loaded active running Avahi mDNS/DNS-SD Stack
  bluetooth.service               loaded active running Bluetooth service
  chrony.service                  loaded active running chrony, an NTP client/server
  colord.service                  loaded active running Manage, Install and Generate Color Profiles
  cron.service                    loaded active running Regular background program processing daemon
  cups-browsed.service            loaded active running Make remote CUPS printers available locally
  cups.service                    loaded active running CUPS Scheduler
  dbus.service                    loaded active running D-Bus System Message Bus
  fwupd.service                   loaded active running Firmware update daemon
  gdm.service                     loaded active running GNOME Display Manager
  ModemManager.service            loaded active running Modem Manager
  networkd-dispatcher.service     loaded active running Dispatcher daemon for systemd-networkd
  NetworkManager.service          loaded active running Network Manager
  polkit.service                  loaded active running Authorization Manager
  power-profiles-daemon.service loaded active running Power Profiles daemon
  rsyslog.service                 loaded active running System Logging Service
  rtkit-daemon.service            loaded active running RealtimeKit Scheduling Policy Service
  snapd.service                   loaded active running Snap Daemon
  switcheroo-control.service      loaded active running Switcheroo Control Proxy service
  systemd-journald.service        loaded active running Journal Service
  systemd-logind.service          loaded active running User Login Management
  systemd-oomd.service            loaded active running Userspace Out-Of-Memory (OOM) Killer
  systemd-resolved.service        loaded active running Network Name Resolution
  systemd-udevd.service           loaded active running Rule-based Manager for Device Events and Files
  udisks2.service                 loaded active running Disk Manager
  unattended-upgrades.service     loaded active running Unattended Upgrades Shutdown
  upower.service                  loaded active running Daemon for power management
  user@1000.service               loaded active running User Manager for UID 1000
  uuidd.service                   loaded active running Daemon for generating UUIDs
  wpa_supplicant.service          loaded active running WPA supplicant
```

This displayed the services currently running in the background on my Ubuntu system.

3. Checking network connections

I used the following commands:

ss -tulnp

ss -tunap

```
maso-prince@maso-prince-VMware-Virtual-Platform:~/CyberOps/Week6$ ss -tulnp
Netid    State     Recv-Q    Send-Q                          Local Address:Port          Peer Address:Port    Process
udp      UNCONN    0         0                                     0.0.0.0:49937              0.0.0.0:*        users:(("python3",pid=4610,
udp      UNCONN    0         0                                     0.0.0.0:5353               0.0.0.0:*
udp      UNCONN    0         0                               192.168.134.128:3702             0.0.0.0:*        users:(("python3",pid=4610,
udp      UNCONN    0         0                               239.255.255.250:3702             0.0.0.0:*        users:(("python3",pid=4610,
udp      UNCONN    0         0                                    127.0.0.54:53               0.0.0.0:*
udp      UNCONN    0         0                                  127.0.0.53%lo:53              0.0.0.0:*
udp      UNCONN    0         0                                    127.0.0.1:323               0.0.0.0:*
udp      UNCONN    0         0                                        *:55981                     *:*        users:(("python3",pid=4610,
udp      UNCONN    0         0                                      [::]:5353                  [::]:*
udp      UNCONN    0         0           [fe80::20c:29ff:fe4a:af9c]%ens33:3702                  [::]:*        users:(("python3",pid=4610,
udp      UNCONN    0         0                          [ff02::c]%ens33:3702                  [::]:*        users:(("python3",pid=4610,
udp      UNCONN    0         0                                    [::1]:323                   [::]:*
tcp      LISTEN    0         4096                             127.0.0.54:53               0.0.0.0:*
tcp      LISTEN    0         4096                           127.0.0.53%lo:53              0.0.0.0:*
tcp      LISTEN    0         4096                             127.0.0.1:631               0.0.0.0:*
tcp      LISTEN    0         4096                               [::1]:631                  [::]:*
maso-prince@maso-prince-VMware-Virtual-Platform:~/CyberOps/Week6$ ss -tunap
Netid    State     Recv-Q    Send-Q                          Local Address:Port          Peer Address:Port    Process
udp      UNCONN    0         0                                     0.0.0.0:49937              0.0.0.0:*        users:(("python3",pid=4610,
udp      UNCONN    0         0                                     0.0.0.0:5353               0.0.0.0:*
udp      UNCONN    0         0                               192.168.134.128:3702             0.0.0.0:*        users:(("python3",pid=4610,
udp      UNCONN    0         0                               239.255.255.250:3702             0.0.0.0:*        users:(("python3",pid=4610,
udp      UNCONN    0         0                                    127.0.0.54:53               0.0.0.0:*
udp      UNCONN    0         0                                  127.0.0.53%lo:53              0.0.0.0:*
udp      ESTAB     0         0                       192.168.134.128%ens33:68          192.168.134.254:67
udp      UNCONN    0         0                                    127.0.0.1:323               0.0.0.0:*
udp      UNCONN    0         0                                        *:55981                     *:*        users:(("python3",pid=4610,
udp      UNCONN    0         0                                      [::]:5353                  [::]:*
udp      UNCONN    0         0           [fe80::20c:29ff:fe4a:af9c]%ens33:3702                  [::]:*        users:(("python3",pid=4610,
udp      UNCONN    0         0                          [ff02::c]%ens33:3702                  [::]:*        users:(("python3",pid=4610,
udp      UNCONN    0         0                                    [::1]:323                   [::]:*
tcp      LISTEN    0         4096                             127.0.0.54:53               0.0.0.0:*
```

These commands helped me check open ports, listening services and network connections on my computer.

4. Checking system logs

I used the following command:

sudo journalctl -n 30

```
maso-prince@maso-prince-VMware-Virtual-Platform:~/CyberOps/Week6$ sudo journalctl -n 30
[sudo: authenticate] Password:
Oct 30 17:59:04 maso-prince-VMware-Virtual-Platform gnome-shell[2751]: DING: DBus interface for Gvfs daemon (org.gtk.vfs.Metadata) is now a
Oct 30 17:59:04 maso-prince-VMware-Virtual-Platform gnome-shell[2751]: DING: DBus interface for Nautilus (org.gnome.Nautilus.FileOperations
Oct 30 17:59:04 maso-prince-VMware-Virtual-Platform gnome-shell[2751]: DING: DBus interface for Nautilus (org.freedesktop.FileManager1) is
Oct 30 17:59:04 maso-prince-VMware-Virtual-Platform gnome-shell[2751]: DING: DBus interface for Switcheroo control (net.hadess.SwitcherooCo
Oct 30 17:59:04 maso-prince-VMware-Virtual-Platform gnome-shell[2751]: Received notification for window. 0 notifications remaining.
Oct 30 17:59:23 maso-prince-VMware-Virtual-Platform systemd[1]: fprintd.service: Deactivated successfully.
Oct 30 17:59:24 maso-prince-VMware-Virtual-Platform kernel: workqueue: css_killed_work_fn hogged CPU for >13333us 7 times, consider switchi
Oct 30 17:59:25 maso-prince-VMware-Virtual-Platform systemd[2436]: ptyxis-spawn-cbe7382c-e40c-45e2-afb4-43a45deed2b2.scope: Consumed 2.253s
Oct 30 17:59:39 maso-prince-VMware-Virtual-Platform systemd[2436]: ptyxis-spawn-56bda55d-7d23-46d9-be5c-b316d10454d7.scope: Consumed 1min 6
Oct 30 18:00:07 maso-prince-VMware-Virtual-Platform systemd[1]: Starting sysstat-collect.service - system activity accounting tool...
Oct 30 18:00:07 maso-prince-VMware-Virtual-Platform systemd[1]: sysstat-collect.service: Deactivated successfully.
Oct 30 18:00:07 maso-prince-VMware-Virtual-Platform systemd[1]: Finished sysstat-collect.service - system activity accounting tool.
Oct 30 18:00:21 maso-prince-VMware-Virtual-Platform systemd[2436]: Started snap.firmware-updater.firmware-notifier.service - Service for sn
Oct 30 18:00:25 maso-prince-VMware-Virtual-Platform kernel: audit: type=1400 audit(1793376025.014:277): apparmor="DENIED" operation="open"
Oct 30 18:00:25 maso-prince-VMware-Virtual-Platform systemd[2436]: snap.firmware-updater.firmware-notifier.service: Consumed 3.447s CPU tim
Oct 30 18:00:49 maso-prince-VMware-Virtual-Platform NetworkManager[965]: <info>  [1793376049.0650] dhcp4 (ens33): state changed new lease,
Oct 30 18:00:49 maso-prince-VMware-Virtual-Platform dbus-daemon[900]: [system] Activating via systemd: service name='org.freedesktop.nm_dis
Oct 30 18:00:49 maso-prince-VMware-Virtual-Platform systemd[1]: Starting NetworkManager-dispatcher.service - Network Manager Script Dispatc
Oct 30 18:00:49 maso-prince-VMware-Virtual-Platform dbus-daemon[900]: [system] Successfully activated service 'org.freedesktop.nm_dispatche
Oct 30 18:00:49 maso-prince-VMware-Virtual-Platform systemd[1]: Started NetworkManager-dispatcher.service - Network Manager Script Dispatch
Oct 30 18:00:59 maso-prince-VMware-Virtual-Platform systemd[1]: NetworkManager-dispatcher.service: Deactivated successfully.
Oct 30 18:01:11 maso-prince-VMware-Virtual-Platform systemd[2436]: Started ptyxis-spawn-6b007189-10df-4286-8999-bb5b13d36985.scope - [syste
Oct 30 18:06:13 maso-prince-VMware-Virtual-Platform systemd[1]: Starting fwupd-refresh.service - Refresh fwupd metadata and update motd...
Oct 30 18:06:14 maso-prince-VMware-Virtual-Platform systemd[1]: fwupd-refresh.service: Deactivated successfully.
Oct 30 18:06:14 maso-prince-VMware-Virtual-Platform systemd[1]: Finished fwupd-refresh.service - Refresh fwupd metadata and update motd.
Oct 30 18:10:01 maso-prince-VMware-Virtual-Platform systemd[1]: Starting sysstat-collect.service - system activity accounting tool...
Oct 30 18:10:01 maso-prince-VMware-Virtual-Platform systemd[1]: sysstat-collect.service: Deactivated successfully.
Oct 30 18:10:01 maso-prince-VMware-Virtual-Platform systemd[1]: Finished sysstat-collect.service - system activity accounting tool.
Oct 30 18:15:01 maso-prince-VMware-Virtual-Platform sudo[8899]: pam_unix(sudo:session): session opened for user root(uid=0) by maso-prince(
Oct 30 18:15:01 maso-prince-VMware-Virtual-Platform sudo[8899]: maso-prince : TTY=/dev/pts/2 ; PWD=/home/maso-prince/CyberOps/Week6 ; USER=
```

This displayed the latest 30 system log entries. I checked the logs for unusual activities and errors.

5. Saving evidence and logs

I saved my practical results using the following commands:

```
ps aux > processes.txt

systemctl list-units --type=service --state=running > services.txt

ss -tulnp > network_connections.txt

sudo journalctl -n 30 > system_logs.txt
```

I also saved my terminal activities in commands.log.

6. Analysis and conclusion

During this practical, I learned how to investigate running processes, services, network connections and system logs using Ubuntu commands.

I observed how these commands provide information about what is happening on my computer. This is important in SOC operations because security analysts monitor host activity to identify suspicious processes, unauthorised connections and possible malware.

I successfully completed the host intrusion analysis practical and saved my evidence. My next step is to improve my ability to identify unusual processes and investigate possible security threats.